Christopher P. Ravosa

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SUMMARY

Software engineer focused on backend systems, real-time distributed services, and high-throughput data processing. Experienced building production APIs, scalable data pipelines, and reliable systems under load.

SKILLS

Languages:	C#, Python, TypeScript, JavaScript, C++, Java
Systems:	Distributed Systems, Real-Time Systems, TCP Networking, REST APIs, WebSockets
Data & Infrastructure:	MongoDB, NoSQL Modeling, Aggregation Pipelines, GCP, AWS
Tools:	Node.js, React Native, Git, CI/CD

PROFESSIONAL EXPERIENCE

Senior Software Engineer - Moku
Mar 2025 - Present

- Built systems to run and manage large-scale, parallel simulations for training and evaluating autonomous agents
- Implemented a runtime configuration system for defining simulation environments and agent behavior, allowing iteration cycles to be **reduced by a period of 60 days**
- Developed a deterministic event and state recording pipeline for debugging, auditing, and replay of system behavior
- Designed modular systems that could be enabled or disabled based on execution context, improving system flexibility
- Created internal tooling including structured logging, debugging dashboards, and configuration management systems to improve developer velocity

Associate Software Engineer - Major League Baseball
Aug 2023 - Mar 2025

- Developed TCP-based matchmaking services and backend APIs supporting real-time user interactions
- Built backend systems to handle high-throughput requests and maintain performance under peak load conditions
- Designed and executed automated stress testing frameworks for backend services, identifying memory leaks and data integrity issues and improving system reliability under sustained load
- Implemented data processing and leaderboard systems handling large player datasets, leveraging MongoDB aggregation pipelines to efficiently query and rank millions of records
- Contributed to content management system enabling configuration of live services without new releases
- Integrated external data sources including MLB StatCast APIs, transforming high-volume, real-world data into interactive systems
- Participated in on-call rotations, diagnosing and resolving production incidents in live systems, including recovery of backend services during outages
- Built pipelines for asset distribution via CDN (GCP), enabling dynamic content delivery while reducing client footprint

Associate Technical Designer - Naughty Dog
Jul 2022 - Jun 2023

- Built networked systems using a host-authoritative architecture, ensuring state consistency across distributed clients
- Implemented synchronization logic resilient to latency, packet loss, and unreliable network conditions
- Designed and tested systems to prevent state divergence and maintain consistency in real-time environments

PROJECTS

Real-Time Climate Monitoring System

- Built an IoT device and fault-tolerant, climate monitoring firmware for an ESP32 microcontroller using ESP-IDF and C
- Implemented bidirectional WebSocket communication between firmware, cloud-hosted backend, and browser clients
- Automated deployment of backend services to Google Cloud Run
- Designed and built a web dashboard for end users to view data and remotely control the IoT device

EDUCATION

- **MSc in Software Development - Concentration in Mobile Computing**
  Marist College - *Sept. 2020 to May 2022*
- **BSc in Computer Science**
  Marist College - *Sept. 2017 to May 2021* ***(GPA 3.87)***